第 14 次作业答案

2020 年4 月10 日

5.26 试求质点组的动量 $\vec{p}$ 和角动量 $\vec{J}$ 的笛卡尔分量所组成的泊松括号

解答：

n 个质点组成的质点组，动量和角动量分量表达式分别为

$$p_x=\sum_{i=1}^{n}p_{ix},\quad p_y=\sum_{i=1}^{n}p_{iy},\quad p_z=\sum_{i=1}^{n}p_{iz}, \tag{1}$$

$$J_x=\sum_{i=1}^{n}(y_ip_{iz}-z_ip_{iy}),\quad J_y=\sum_{i=1}^{n}(z_ip_{ix}-x_ip_{iz}),\quad J_z=\sum_{i=1}^{n}(x_ip_{iy}-y_ip_{ix}). \tag{2}$$

系统的正则变量为

$$x_i,y_i,z_i,p_{ix},p_{iy},p_{iz},\quad (i=1,2,\ldots,n). \tag{3}$$

两函数的泊松括号为

$$[A,B]=\sum_{i=1}^{n}\left[\left(\frac{\partial A}{\partial x_i}\frac{\partial B}{\partial p_{ix}}-\frac{\partial A}{\partial p_{ix}}\frac{\partial B}{\partial x_i}\right)+\left(\frac{\partial A}{\partial y_i}\frac{\partial B}{\partial p_{iy}}-\frac{\partial A}{\partial p_{iy}}\frac{\partial B}{\partial y_i}\right)+\left(\frac{\partial A}{\partial z_i}\frac{\partial B}{\partial p_{iz}}-\frac{\partial A}{\partial p_{iz}}\frac{\partial B}{\partial z_i}\right)\right] \tag{4}$$

根据 p_α 和 J_α 的表达式，易得

$$\frac{\partial J_x}{\partial x_i}=\frac{\partial J_y}{\partial y_i}=\frac{\partial J_z}{\partial z_i}=\frac{\partial J_x}{\partial p_{ix}}=\frac{\partial J_y}{\partial p_{iy}}=\frac{\partial J_z}{\partial p_{iz}}=0, \tag{5}$$

$$\frac{\partial J_x}{\partial y_i}=-\frac{\partial J_y}{\partial x_i}=p_{iz},\quad -\frac{\partial J_x}{\partial z_i}=\frac{\partial J_z}{\partial x_i}=p_{iy},\quad \frac{\partial J_y}{\partial z_i}=-\frac{\partial J_z}{\partial y_i}=p_{ix}, \tag{6}$$

$$\frac{\partial p_x}{\partial p_{ix}}=\frac{\partial p_y}{\partial p_{iy}}=\frac{\partial p_z}{\partial p_{iz}}=1, \tag{7}$$

$$\frac{\partial p_x}{\partial p_{iy}}=\frac{\partial p_x}{\partial p_{iz}}=\frac{\partial p_y}{\partial p_{ix}}=\frac{\partial p_y}{\partial p_{iz}}=\frac{\partial p_z}{\partial p_{ix}}=\frac{\partial p_z}{\partial p_{iy}}=0 \tag{8}$$

直接应用泊松括号的定义，因为正则变量彼此独立，易见上式中仅有小括号中第一项有贡献，因此角动量与动量分量的泊松括号分别为：

$$[J_x,p_x]=\sum_{i=1}^{n}\left(\frac{\partial J_x}{\partial x_i}\frac{\partial p_x}{\partial p_{ix}}+\frac{\partial J_x}{\partial y_i}\frac{\partial p_x}{\partial p_{iy}}+\frac{\partial J_x}{\partial z_i}\frac{\partial p_x}{\partial p_{iz}}\right)=0, \tag{9}$$

$$[J_x,p_y]=\sum_{i=1}^{n}\left(\frac{\partial J_x}{\partial x_i}\frac{\partial p_y}{\partial p_{ix}}+\frac{\partial J_x}{\partial y_i}\frac{\partial p_y}{\partial p_{iy}}+\frac{\partial J_x}{\partial z_i}\frac{\partial p_y}{\partial p_{iz}}\right)=\sum_{i=1}^{n}p_{iz}=p_z, \tag{10}$$

$$[J_x,p_z]=\sum_{i=1}^{n}\left(\frac{\partial J_x}{\partial x_i}\frac{\partial p_z}{\partial p_{ix}}+\frac{\partial J_x}{\partial y_i}\frac{\partial p_z}{\partial p_{iy}}+\frac{\partial J_x}{\partial z_i}\frac{\partial p_z}{\partial p_{iz}}\right)=\sum_{i=1}^{n}(-p_{iy})=-p_y, \tag{11}$$

$$[J_y,p_x]=\sum_{i=1}^{n}\left(\frac{\partial J_y}{\partial x_i}\frac{\partial p_x}{\partial p_{ix}}+\frac{\partial J_y}{\partial y_i}\frac{\partial p_x}{\partial p_{iy}}+\frac{\partial J_y}{\partial z_i}\frac{\partial p_x}{\partial p_{iz}}\right)=\sum_{i=1}^{n}(-p_{iz})=-p_z, \tag{12}$$

$$[J_y,p_y]=\sum_{i=1}^{n}\left(\frac{\partial J_y}{\partial x_i}\frac{\partial p_y}{\partial p_{ix}}+\frac{\partial J_y}{\partial y_i}\frac{\partial p_y}{\partial p_{iy}}+\frac{\partial J_y}{\partial z_i}\frac{\partial p_y}{\partial p_{iz}}\right)=0, \tag{13}$$

$$[J_y,p_z]=\sum_{i=1}^{n}\left(\frac{\partial J_y}{\partial x_i}\frac{\partial p_z}{\partial p_{ix}}+\frac{\partial J_y}{\partial y_i}\frac{\partial p_z}{\partial p_{iy}}+\frac{\partial J_y}{\partial z_i}\frac{\partial p_z}{\partial p_{iz}}\right)=\sum_{i=1}^{n}p_{ix}=p_x, \tag{14}$$

$$[J_z,p_x]=\sum_{i=1}^{n}\left(\frac{\partial J_z}{\partial x_i}\frac{\partial p_x}{\partial p_{ix}}+\frac{\partial J_z}{\partial y_i}\frac{\partial p_x}{\partial p_{iy}}+\frac{\partial J_z}{\partial z_i}\frac{\partial p_x}{\partial p_{iz}}\right)=\sum_{i=1}^{n}p_{iy}=p_y, \tag{15}$$

$$[J_z,p_y]=\sum_{i=1}^{n}\left(\frac{\partial J_z}{\partial x_i}\frac{\partial p_y}{\partial p_{ix}}+\frac{\partial J_z}{\partial y_i}\frac{\partial p_y}{\partial p_{iy}}+\frac{\partial J_z}{\partial z_i}\frac{\partial p_y}{\partial p_{iz}}\right)=\sum_{i=1}^{n}(-p_{ix})=-p_x, \tag{16}$$

$$[J_z,p_z]=\sum_{i=1}^{n}\left(\frac{\partial J_z}{\partial x_i}\frac{\partial p_z}{\partial p_{ix}}+\frac{\partial J_z}{\partial y_i}\frac{\partial p_z}{\partial p_{iy}}+\frac{\partial J_z}{\partial z_i}\frac{\partial p_z}{\partial p_{iz}}\right)=0. \tag{17}$$

总结一下，角动量和动量分量的泊松括号为

$$[J_x,p_x]=0,\quad [J_x,p_y]=p_z,\quad [J_x,p_z]=-p_y, \tag{18}$$

$$[J_y,p_x]=-p_z,\quad [J_y,p_y]=0,\quad [J_y,p_z]=p_x, \tag{19}$$

$$[J_z,p_x]=p_y,\quad [J_z,p_y]=-p_x,\quad [J_z,p_z]=0. \tag{20}$$

可以简写为

$$[J_i,p_j]=\sum_{k=1}^{3}\epsilon^{ijk}p_k. \tag{21}$$

上式中

$$\epsilon^{ijk}=\begin{cases}1 & (ijk\text{ 是 }123\text{ 的偶排列})\\ -1 & (ijk\text{ 是 }123\text{ 的奇排列})\\ 0 & (\text{其他情况})\end{cases} \tag{22}$$

5.34 利用下列关系把变数从 p,q 变为 P,Q:

$$q=\varphi_1(P,Q),\quad p=\varphi_2(P,Q), \tag{23}$$

则当

$$\frac{\partial(q,p)}{\partial(Q,P)}=1 \tag{24}$$

时，这种变换是一正则变换。试证明之。

解法一：利用全微分条件 $\nabla\times\vec{F}=0$

正则变换的条件为 $p\mathrm{d}q$ 和 $-P\mathrm{d}Q$ 对 α 求和后构成一全微分

$$p\mathrm{d}q-P\mathrm{d}Q+(H^*-H)\mathrm{d}t=\mathrm{d}U. \tag{25}$$

左方构成全微分，则必有

$$p=\frac{\partial U}{\partial q},\quad P=-\frac{\partial U}{\partial Q}\quad\Longrightarrow\quad \frac{\partial p}{\partial Q}+\frac{\partial P}{\partial q}=0 \tag{26}$$

现在问题归结于检验上式是否成立。

根据前边 p,q 与 P,Q 的变换关系，

$$\mathrm{d}q=\frac{\partial\varphi_1}{\partial P}\mathrm{d}P+\frac{\partial\varphi_1}{\partial Q}\mathrm{d}Q \tag{27}$$

$$\mathrm{d}p=\frac{\partial\varphi_2}{\partial P}\mathrm{d}P+\frac{\partial\varphi_2}{\partial Q}\mathrm{d}Q. \tag{28}$$

两式消去 $\mathrm{d}Q$，易得

$$\frac{\partial\varphi_2}{\partial Q}\mathrm{d}q-\frac{\partial\varphi_1}{\partial Q}\mathrm{d}p=-\frac{\partial(q,p)}{\partial(Q,P)}\mathrm{d}P. \tag{29}$$

易得

$$\frac{\partial P}{\partial q}=-\left[\frac{\partial(q,p)}{\partial(Q,P)}\right]^{-1}\frac{\partial\varphi_2}{\partial Q}. \tag{30}$$

代入上式，易得

$$\frac{\partial p}{\partial Q}+\frac{\partial P}{\partial q}=\frac{\partial\varphi_2}{\partial Q}\left\{1-\left[\frac{\partial(q,p)}{\partial(Q,P)}\right]^{-1}\right\}=0. \tag{31}$$

最后等号利用了题目给的条件 $\dfrac{\partial(q,p)}{\partial(Q,P)}=1.$

解法二：直接构造全微分

正则变换的条件为 $p\mathrm{d}q$ 和 $-P\mathrm{d}Q$ 对 α 求和后构成一全微分

$$p\mathrm{d}q-P\mathrm{d}Q+(H^*-H)\mathrm{d}t=\mathrm{d}U. \tag{32}$$

要想证明正则变换，只需证明前两项构成全微分即可。

$$p\mathrm{d}q-P\mathrm{d}Q=\varphi_2\mathrm{d}\varphi_1-P\mathrm{d}Q=\left(\varphi_2\frac{\partial\varphi_1}{\partial Q}-P\right)\mathrm{d}Q+\varphi_2\frac{\partial\varphi_1}{\partial P}\mathrm{d}P=f_1\mathrm{d}Q+f_2\mathrm{d}P. \tag{33}$$

上式为全微分的充要条件为 $\dfrac{\partial f_1}{\partial P}-\dfrac{\partial f_2}{\partial Q}=0.$下面开始验证之：

$$\frac{\partial f_1}{\partial P}-\frac{\partial f_2}{\partial Q}=\left(\frac{\partial\varphi_2}{\partial P}\frac{\partial\varphi_1}{\partial Q}-\varphi_2\frac{\partial^2\varphi_1}{\partial Q\partial P}-1\right)-\left(\frac{\partial\varphi_2}{\partial Q}\frac{\partial\varphi_1}{\partial P}+\varphi_2\frac{\partial^2\varphi_1}{\partial Q\partial P}\right) \tag{34}$$

$$=\frac{\partial(q,p)}{\partial(Q,P)}-1=0. \tag{35}$$

证毕。

5.35 试利用正则变换，由正则方程求竖直上抛的物体的运动规律。已知本问题的母函数 $U=mg\left(\dfrac{1}{6}gQ^3+qQ\right)$，式中 q 为确定物体位置的广义坐标，Q 为变换后的新的广义坐标，g 为重力加速度。

解答：

竖直上抛运动，物体的拉格朗日函数为

$$L = T - V = \frac{1}{2}m\dot{q}^2 - mgq. \tag{36}$$

应用勒让德变换，易得

$$p = \frac{\partial L}{\partial \dot{q}} = m\dot{q} \tag{37}$$

$$H = p\dot{q} - L = \frac{p^2}{2m} + mgq. \tag{38}$$

易见母函数依赖于 (q, Q)，正则变换满足的关系式为

$$p\mathrm{d}q - P\mathrm{d}Q + (H^* - H)\mathrm{d}t = \mathrm{d}U, \quad U \equiv U(q, Q, t). \tag{39}$$

根据以上关系，易得

$$p = \frac{\partial U}{\partial q} = mg\mathrm{Q}, \tag{40}$$

$$P = -\frac{\partial U}{\partial Q} = mg\left(\frac{1}{2}g\mathrm{Q}^2 + q\right) \quad \Longrightarrow \quad q = \frac{P}{mg} - \frac{1}{2}g\mathrm{Q}^2 \tag{41}$$

$$\tag{42}$$

由此得到变换后的哈密顿函数为

$$H^* = H + \frac{\partial U}{\partial t} = H \tag{43}$$

$$= \frac{(mg\mathrm{Q})^2}{2m} + mg\left(\frac{P}{mg} - \frac{1}{2}g\mathrm{Q}^2\right) \tag{44}$$

$$= P. \tag{45}$$

哈密顿方程给出

$$\dot{\mathrm{Q}} = \frac{\partial H}{\partial P} = 1 \quad \Longrightarrow \quad \mathrm{Q} = t + C_1, \tag{46}$$

$$\dot{P} = -\frac{\partial H}{\partial \mathrm{Q}} = 0 \quad \Longrightarrow \quad P = C_2 \tag{47}$$

将 Q, P 代回 q 的表达式中，易得

$$q = \frac{C_2}{mg} - \frac{1}{2}g(t + C_1)^2. \tag{48}$$

代入初始条件 q_0, v_0，即

$$q_0 = \frac{C_2}{mg} - \frac{1}{2}gC_1^2, \quad v_0 = -gC_1 \tag{49}$$

$$\Longrightarrow \quad C_1 = -\frac{v_0}{g}, \quad \frac{C_2}{mg} = q_0 + \frac{v_0^2}{2g}, \tag{50}$$

$$\Longrightarrow \quad q = q_0 + v_0 t - \frac{1}{2}gt^2. \tag{51}$$

5.36 试求质点在势场

$$V=\frac{\alpha}{r^2}-\frac{Fz}{r^3} \tag{52}$$

中运动的主函数 S，式中 α 及 F 为常数。

解答：

选取球坐标系，系统的哈密顿函数为

$$H=T+V=\frac{p_r^2}{2m}+\frac{p_\theta^2}{2mr^2}+\frac{p_\varphi^2}{2mr^2\sin^2\theta}+\frac{\alpha}{r^2}-\frac{F}{r^2}\cos\theta. \tag{53}$$

因为 $\dfrac{\partial H}{\partial t}=0$，故主函数可以写为

$$S=-Et+W \tag{54}$$

其中 W 满足如下的方程

$$H\left(r,\theta,\varphi;\frac{\partial W}{\partial r},\frac{\partial W}{\partial \theta},\frac{\partial W}{\partial \varphi}\right)=E \tag{55}$$

上式中 $p_i=\dfrac{\partial W}{\partial q_i}$.

将哈密顿函数的形式代入上式，稍加整理之后得到

$$\frac{1}{2m}\left(\frac{\partial W}{\partial r}\right)^2+\frac{\alpha}{r^2}+\frac{1}{2mr^2}\left[\left(\frac{\partial W}{\partial \theta}\right)^2-2mF\cos\theta\right]+\frac{1}{2mr^2\sin^2\theta}\left(\frac{\partial W}{\partial \varphi}\right)^2=E. \tag{56}$$

显然 φ 是循环坐标，对应守恒量 p_φ，所求的解具有如下形式

$$W=p_\varphi\varphi+W_1(r)+W_2(\theta). \tag{57}$$

将其代入上式，易见运动方程变为

$$\frac{1}{2m}\left(\frac{\mathrm{d}W_1}{\mathrm{d}r}\right)^2+\frac{\alpha}{r^2}+\frac{1}{2mr^2}\left[\left(\frac{\mathrm{d}W_2}{\mathrm{d}\theta}\right)^2-2mF\cos\theta+\frac{p_\varphi^2}{\sin^2\theta}\right]=E. \tag{58}$$

上式中对 θ 依赖的部分可以分离出来，故其应该等于常数，上述方程可以分离变量为

$$\left(\frac{\mathrm{d}W_2}{\mathrm{d}\theta}\right)^2-2mF\cos\theta+\frac{p_\varphi^2}{\sin^2\theta}=\alpha_2, \tag{59}$$

$$\frac{1}{2m}\left(\frac{\mathrm{d}W_1}{\mathrm{d}r}\right)^2+\frac{\alpha}{r^2}+\frac{\alpha_2}{2mr^2}=E. \tag{60}$$

由上式可以得到 W_1 和 W_2 的微分形式，

$$\frac{\mathrm{d}W_1}{\mathrm{d}r}=\sqrt{2m\left(E-\frac{\alpha}{r^2}\right)-\frac{\alpha_2}{r^2}}, \tag{61}$$

$$\frac{\mathrm{d}W_2}{\mathrm{d}\theta}=\sqrt{\alpha_2+2mF\cos\theta-\frac{p_\varphi^2}{\sin^2\theta}}. \tag{62}$$

主函数表达式为

$$S=-Et+p_\varphi\varphi+\int\sqrt{2m\left(E-\frac{\alpha}{r^2}\right)-\frac{\alpha_2}{r^2}}\mathrm{d}r+\int\sqrt{\alpha_2+2mF\cos\theta-\frac{p_\varphi^2}{\sin^2\theta}}\mathrm{d}\theta. \tag{63}$$

5.39 试用哈-雅方程求解行星绕太阳运动时的轨道方程。

解答：

行星绕太阳运动，根据角动量守恒，行星在平面内运动，故可以在极坐标系中处理。行星的哈密顿函数为

$$H = T + V = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} - \frac{mk^2}{r}. \tag{64}$$

显然系统的哈密顿函数不显含时间 t，$\dfrac{\partial H}{\partial t} = 0$，故主函数可以写为

$$S = -Et + W, \tag{65}$$

其中哈密顿-雅可比特性函数 W 满足

$$\frac{1}{2m}\left(\frac{\partial W}{\partial r}\right)^2 + \frac{1}{2mr^2}\left(\frac{\partial W}{\partial \theta}\right)^2 - \frac{mk^2}{r} = E \tag{66}$$

因为 θ 是循环坐标，故对应守恒律 p_θ，因此

$$W = p_\theta \theta + W_1(r) \tag{67}$$

代入上式，易得

$$\frac{1}{2m}\left(\frac{\mathrm{d}W_1}{\mathrm{d}r}\right)^2 + \frac{p_\theta^2}{2mr^2} - \frac{mk^2}{r} = E. \tag{68}$$

$$\Longrightarrow \qquad \frac{\mathrm{d}W_1}{\mathrm{d}r} = \sqrt{2mE + \frac{2m^2k^2}{r} - \frac{p_\theta^2}{r^2}}. \tag{69}$$

系统的主函数因此为

$$S = -Et + p_\theta \theta + \int \sqrt{2mE + \frac{2m^2k^2}{r} - \frac{p_\theta^2}{r^2}}\,\mathrm{d}r \tag{70}$$

求行星的轨道方程，需要用到(5.9.12)中的第 3 式，即

$$\frac{\partial W}{\partial \alpha_i} = \beta_i \tag{71}$$

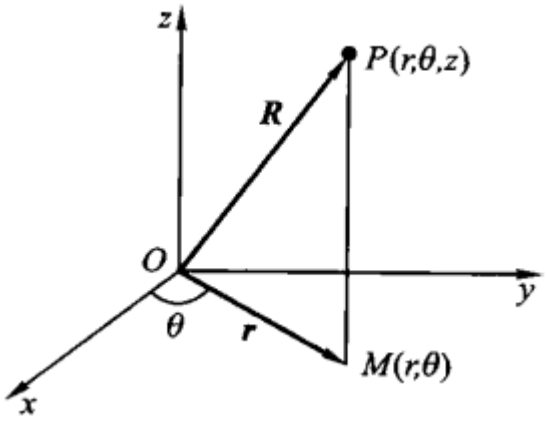
上式中 p_θ 即为对应的 α_i，故方程变为

$$\theta - \int \frac{p_\theta / r^2}{\sqrt{2mE + \frac{2m^2k^2}{r} - \frac{p_\theta^2}{r^2}}}\,\mathrm{d}r = C_0 \tag{72}$$

将 C_0 吸收到不定积分中，易得轨道方程为

$$\theta = \int \frac{p_\theta / r^2}{\sqrt{2mE + \frac{2m^2k^2}{r} - \frac{p_\theta^2}{r^2}}}\,\mathrm{d}r. \tag{73}$$

此即 §1.9 中轨道方程的形式，其中 $p_\theta = J$，为行星的轨道角动量。



5.41 试求质点在库仑场和均匀场

$$V=\frac{\alpha}{R}-Fz \tag{74}$$

的合成场中运动是的主函数 S，以抛物线坐标 ξ,η,θ 表示，式中 α 及 F 是常数，而 $R=\sqrt{r^2+z^2}$.

解答：

根据教材 §5.9 的内容，抛物线坐标 ξ,η 为柱坐标 r,z 的变换

$$z=\frac{1}{2}(\xi-\eta),\quad r=\sqrt{\xi\eta}. \tag{75}$$

其中 ξ,η 的取值范围为 $[0,\infty)$. 在抛物线坐标中，系统的哈密顿函数(5.9.24)式，

$$H=\frac{2}{m}\frac{\xi p_\xi^2+\eta p_\eta^2}{\xi+\eta}+\frac{p_\theta^2}{2m\xi\eta}+V(\xi,\eta,\theta). \tag{76}$$

根据 z 和 r 的表达式，易见

$$R=\sqrt{r^2+z^2}=\sqrt{\frac{1}{4}(\xi-\eta)^2+\xi\eta}=\frac{\xi+\eta}{2}. \tag{77}$$

代入势函数表达式，易得哈密顿函数形式为

$$H=\frac{2}{m}\frac{\xi p_\xi^2+\eta p_\eta^2}{\xi+\eta}+\frac{p_\theta^2}{2m\xi\eta}+\frac{2\alpha}{\xi+\eta}-\frac{F}{2}(\xi-\eta). \tag{78}$$

因为哈密顿函数不显含 t，主函数形式为

$$S=-Et+W \tag{79}$$

其中 W 满足如下方程

$$\frac{2}{m}\frac{\xi}{\xi+\eta}\left(\frac{\partial W}{\partial\xi}\right)^2+\frac{2}{m}\frac{\eta}{\xi+\eta}\left(\frac{\partial W}{\partial\eta}\right)^2+\frac{1}{2m\xi\eta}\left(\frac{\partial W}{\partial\theta}\right)^2+\frac{2\alpha}{\xi+\eta}-\frac{F}{2}(\xi-\eta)=E. \tag{80}$$

因为 θ 是系统的循环坐标，存在守恒律 p_θ，特性函数 W 可以写为

$$W=p_\theta\theta+W_1(\xi)+W_2(\eta). \tag{81}$$

将 W 满足的方程乘以 $\xi+\eta$，代入上式，易得

$$\frac{2}{m}\xi\left(\frac{\mathrm{d}W_1}{\mathrm{d}\xi}\right)^2+\frac{2}{m}\eta\left(\frac{\mathrm{d}W_2}{\mathrm{d}\eta}\right)^2+\frac{p_\theta^2}{2m}\left(\frac{1}{\xi}+\frac{1}{\eta}\right)+2\alpha-\frac{F}{2}(\xi^2-\eta^2)=(\xi+\eta)E. \tag{82}$$

很容易将上式中对 ξ 和 η 的依赖分开，故两式都等于常数，有

$$2\xi\left(\frac{\mathrm{d}W_1}{\mathrm{d}\xi}\right)^2+\frac{p_\theta^2}{2\xi}+m\alpha-\frac{mF}{2}\xi^2-m\xi E=\alpha_2 \tag{83}$$

$$2\eta\left(\frac{\mathrm{d}W_2}{\mathrm{d}\eta}\right)^2+\frac{p_\theta^2}{2\eta}+m\alpha+\frac{mF}{2}\eta^2-m\eta E=-\alpha_2. \tag{84}$$

求得 $\frac{\mathrm{d}W_1}{\mathrm{d}\xi}$ 和 $\frac{\mathrm{d}W_2}{\mathrm{d}\eta}$ 之后，易得主函数的形式为

$$S=-Et+p_\theta\theta+\int\sqrt{\frac{mE}{2}+\frac{\alpha_2}{2\xi}+\frac{mF\xi}{4}-\frac{m\alpha}{2\xi}-\frac{p_\theta^2}{4\xi^2}}\mathrm{d}\xi \tag{85}$$

$$+\int\sqrt{\frac{mE}{2}-\frac{\alpha_2}{2\eta}+\frac{mF\eta}{4}-\frac{m\alpha}{2\eta}-\frac{p_\theta^2}{4\eta^2}}\mathrm{d}\eta. \tag{86}$$